Detection of Spatially Oriented Fusarium Head Blight Spikes in Wheat Using UAV-Based Remote Sensing Imaging

Nisar Ali, Muhib Ullah, *Member, IEEE*, Abdul Bais, *Senior Member, IEEE*, Samia Berraies, Yuefeng Ruan, Richard D. Cuthbert, and Jatinder S. Sangha

***Abstract*—Fusarium head blight (FHB) poses a critical threat to global wheat production, compromising yield and quality through mycotoxin contamination. Traditional methods, such as laboratory assays and visual inspections, are labor-intensive, costly, and impractical for large-scale monitoring. Although remote sensing based on unmanned aerial vehicles (UAVs) offers a promising alternative, existing approaches using axis-aligned bounding boxes (AABBs) struggle with arbitrary orientations, large aspect ratios, and dense wheat spike arrangements, limiting detection accuracy. To address these challenges, we introduce a UAV-based remote sensing framework for detecting oriented FHB-infected wheat spikes under real field conditions. We propose a novel modified hybrid YOLO architecture for detecting oriented bounding boxes (OBBs) that integrates the efficiency of YOLOv10 with YOLOv11's orientation-aware detection capabilities. To enhance lesion visibility in UAV images, we utilize pixel-level cubic power stretching, which amplifies the contrast of FHB-infected regions while normalizing overexposed backgrounds. Experimental results demonstrate state-of-the-art performance; for example, our Hybrid-YOLO-*x* achieves 93.8% mAP@0.5, surpassing YOLOv8x and YOLOv11x by 3.1% and 1.3%, respectively. It reduces parameters by 30.4% (versus YOLOv8x) and 17.7% (versus YOLOv11x) while lowering GFLOPs by 28.3% (versus YOLOv8x) and 7.1% (versus YOLOv11x). The model also achieves 67.7% mAP @ 0.5–0.95, outperforming YOLOv8x by 4.3% and YOLOv11x by 0.6%. Our hybrid model also outperforms other state-of-the-art OBB detectors, such as S^2ANet, ROI-Transformer, and R^2CNN, utilizing 12.8% fewer parameters and 5.4% lower computational cost.**

***Index Terms*—Detection, disease, Fusarium head blight (FHB), oriented bounding box (OBB), precision agriculture, remote sensing, unmanned aerial vehicle (UAV), wheat.**

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I. Introduction

FUSARIUM head blight (FHB) is caused by several species of the genus Fusarium, infecting cereals including wheat, barley, oats, and other small grain crops [1], [2]. Worldwide, FHB plays a significant role in influencing the production of wheat, which is the primary staple crop for one-third of the global population [3], [4]. Within different wheat types, durum is more susceptible to FHB than bread wheat [5]. Durum is currently Canada's second-largest seeded wheat type and produced 5.9 million tons of grain during the 2024 crop season [6]. It is also Canada's largest exported wheat class to the world market, playing a critical role in the Canadian economy [7]. FHB infection in durum could impact grade, pasta-making quality, and marketability due to the presence of mycotoxins, especially deoxynivalenol (DON), that pose severe health risks to humans and livestock [8]. The frequency and severity of FHB outbreaks in some wheat-growing regions of the world have increased in recent years, driven by factors such as climate change, crop management practices, and increased virulence of the pathogen [9]. Improving FHB resistance in durum through breeding is, therefore, a key, environmentally benign step to protect durum yield and quality.

FHB resistance is a primary focus of durum breeding programs in Canada and other parts of the world, such as China, USA, and South America [10]. Improving durum resistance to FHB is, however, a major challenge in practicality, requiring a thorough assessment of untapped durum germplasm through a large-scale field screening and laboratory analysis before selecting parents for use in crossing plans [11], [12]. Traditional assessment of FHB resistance in durum breeding relies on labor-intensive disease nurseries and daily post-anthesis field inspections by trained personnel to monitor infection, progression, and severity [13]. This process involves manually rating a large number of germplasm lines in replicated field plots and visits to multiple field sites for collecting disease incidence and severity notes from the representative field plots. Thereafter, FHB-infected spikes are collected from each plot, separating Fusarium-damaged kernels (FDKs) to determine DON mycotoxin [14] in the grain through techniques such as high-performance liquid chromatography [15] and enzyme-linked immunosorbent assays [16]. Subsequently, the level of

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FHB resistance is assigned to individual lines compared to the check varieties. All these activities are intensive, resource demanding and critically time-dependent to make decisions on assigning FHB resistance to the germplasm, delaying the development of durum cultivars with strong FHB resistance [17]. Improving methods for FHB assessment and detection using automated tools will facilitate the high-throughput screening of FHB resistance in wheat.

New technologies have been playing an important role in the detection of various plant pathogens, helping plant pathologists to evaluate the crop resistance [18]. Technologies such as remote sensing [19], [20] are invaluable in fields, including national security [21], urban planning [22], resource surveys [23], and agriculture [24]. Aerial imaging-based remote sensing provides a rapid and nondestructive method for monitoring and identifying FHB in wheat crops grown in a commercial area. Current applications of remote sensing are oriented toward evaluating damaged wheat kernels in laboratory settings [25], [26], analyzing individual plants or canopies affected by FHB [27], and monitoring disease spread at regional scales [28]. Despite its potential, remote sensing approaches face significant challenges, including delayed detection in laboratory settings, limited spatial distribution data in single-plant studies, and interference from cloud cover in optical satellite imagery, particularly during peak FHB outbreak periods [29].

The development of unmanned aerial vehicle (UAV) technology has improved the phenotyping capacity in agriculture, making it a highly efficient and increasingly popular tool for crop classification, growth monitoring, and pest identification [30], [31], [32]. Unlike ground-based or satellite platforms, UAVs are flexible and cost-effective [33], [34]. UAV platforms support various imaging technologies, including red, green, and blue (RGB), near-infrared, multispectral, hyperspectral, and thermal sensors, each offering unique capabilities for detection and management of plant diseases [35], [36]. This is possible since UAVs can quickly cover vast farmland areas, capturing high-resolution images and data that enable comprehensive and accurate disease monitoring [32]. Flying at low altitudes further improves the application of UAVs, which can capture details in crop images that reveal disease-induced changes in crop appearance, including alterations in color, pigmentation, chlorophyll content, shape, and moisture levels [37], [38]. Processing UAV-captured images with machine learning algorithms helped to identify and measure disease symptoms, track disease progression, and create prescription maps [39], enabling precise and efficient disease control in agricultural fields [40]. Application of UAV-based wheat phenotyping thus holds potential to assist the FHB assessment.

In recent years, deep learning (DL) has significantly advanced crop disease classification and severity assessment by eliminating the need for manual feature engineering and enabling end-to-end image analysis pipelines. Within the wheat domain, Fernandez-Campos et al. [41] utilized convolutional neural networks to evaluate wheat blast severity across multiple datasets. Su et al. [42] proposed a dual Mask-RCNN approach to segment wheat spikes and lesion areas independently, enabling accurate FHB severity estimation based on lesion-to-spike area ratios and achieving a recognition accuracy of 77%. Mi et al. [43] further refined disease severity classification using a CdenseNet architecture with convolutional block attention modules and DenseBlocks, reaching 97.9% accuracy for wheat stripe rust severity. Moreover, the integration of DL with UAV-based imaging has enhanced high-throughput crop monitoring and timely disease intervention. Xiao et al. [44] improved FHB detection in wheat by optimizing texture feature extraction from UAV imagery using the gray-level co-occurrence matrix. Kerkech et al. [45] demonstrated the power of multimodal UAV data by combining visible and infrared imagery for accurate grape disease detection, achieving over 87% accuracy. Similarly, Bao et al. [46] applied a dual-dimensional mixed attention YOLO model on UAV-captured images for tea leaf blight detection. These advancements underscore the potential of UAVs paired with DL to drive precision agriculture by facilitating early, accurate crop disease detection, which is key for the assessment of FHB in a disease nursery.

Despite significant advances in UAV-based crop monitoring, detecting FHB in wheat using remote sensing presents unique challenges. Conventional object detection approaches primarily utilize axis-aligned bounding boxes (AABBs), which often include unnecessary background areas and reduce annotation precision. This limitation is problematic for elongated objects, such as wheat spikes, which appear in various orientations depending on the imaging angles and natural growth patterns. Furthermore, the quality of UAV-acquired images is influenced by sensor characteristics, flight altitude, and environmental conditions, particularly high illumination that can cause overexposure and detail loss in FHB-infected areas. The large ground sample distances associated with UAV remote sensing result in lower resolution images with complex backgrounds, further complicating accurate detection and subsequently disease assessment.

However, as shown in Fig. 1, there are three major difficulties specifically related to detecting FHB-infected wheat spikes in the field.

1) *Arbitrary Orientations:* AABB cannot reflect the real sizes and aspect ratios of wheat spikes with various orientations, as they grow in different directions within a field.
2) *Large Aspect Ratios:* Wheat spikes typically have elongated shapes with large aspect ratios. Significant background areas are included when using AABB, reducing the signal-to-noise ratio for disease detection.
3) *Dense Arrangements:* In mature wheat fields, spikes often appear in dense arrangements. AABBs frequently overlap in these areas, making it difficult to distinguish individual spikes and accurately assess disease status.

Oriented bounding boxes (OBBs) offer an alternative for detecting FHB-infected wheat spikes in UAV imagery. Unlike standard AABBs, OBBs align with the actual shape and orientation of wheat spikes rather than constraining them to horizontal or vertical axes. This approach provides a more precise representation of objects with arbitrary orientations, ensuring better localization of FHB-infected spikes and thereby improving the accuracy of the detection model. The

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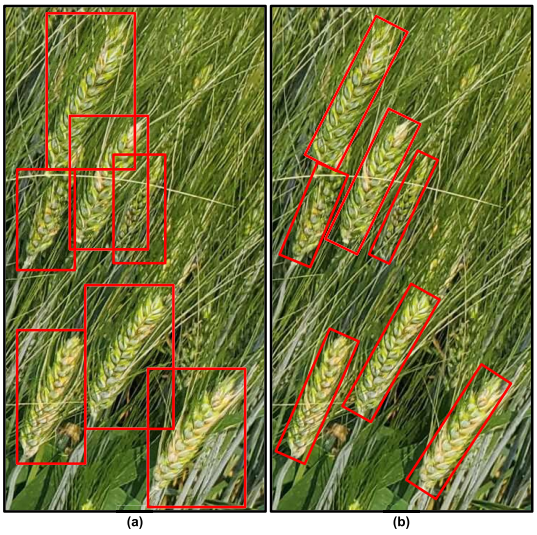


Fig. 1. Comparison of AABB and OBB for detecting FHB wheat spikes. (a) AABB method, where the boxes are aligned with the horizontal and vertical axes, potentially including extraneous background areas and misalignment. (b) OBB process, where the boxes are rotated to align with the spike's true orientation, enhancing localization precision and reducing background noise.

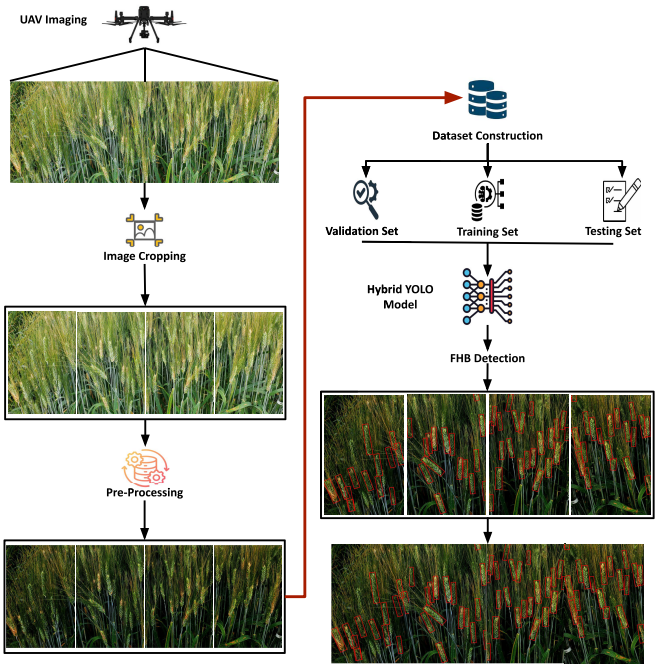


Fig. 2. Complete workflow of the proposed approach for detecting oriented FHB wheat spikes using UAVs.

OBB approach offers several key advantages over AABB for FHB detection.

1) OBBs minimize the inclusion of background pixels by tightly enclosing the wheat spike along its natural orientation, improving the signal-to-noise ratio for disease detection.
2) They provide better localization of FHB-infected areas, enabling more accurate assessment of disease severity and distribution.
3) They are particularly effective for elongated objects like wheat spikes, which often appear in various orientations in UAV imagery due to plant growth patterns and environmental factors.
4) They enhance the ability of detection models to differentiate between healthy and diseased spikes by focusing more precisely on the infected tissue.

Detecting arbitrarily oriented objects, such as FHB wheat spikes, requires an additional angular parameter beyond traditional AABB. Modern OBB detection methods address this issue through angle regression strategies, including direct continuous regression, discretized angle classification, and multistage refinement, which handle angular periodicity and boundary discontinuities [47], [48]. To better optimize overlaps between rotated boxes, rotation-invariant IoU metrics, such as Rotated IoU and Skew IoU, have been introduced to improve alignment accuracy during training [49]. Furthermore, orientation-specific loss functions, like smooth angle loss and modulated rotation loss, enhance model stability and precision in dense scenes [50], [51].

In this article, we introduce a novel modified hybrid YOLO architecture that addresses previous challenges in detecting spatially oriented FHB-infected wheat spikes using UAV-based remote sensing imaging. To our knowledge, this is the first study to implement UAV-based technology for detecting spatially oriented FHB-infected wheat spikes under field conditions, which will open further avenues for assessing FHB disease in the field and help evaluate disease resistance in wheat. Our key contributions include the following.

1) Introducing a UAV-based remote sensing approach for detecting spatially oriented FHB-infected wheat spikes under real field conditions.
2) Developing a novel modified hybrid YOLO model that integrates YOLOv10's efficiency through compact inverted blocks (CIBs) and spatial–channel decoupled downsampling (SCDown) with YOLOv11's orientation-aware detection via C3k2 blocks and depthwise convolutions (DWConvs) heads.
3) Developing a UAV-based dataset of FHB-infected wheat spikes under real field conditions, incorporating oriented annotations to advance research in oriented object detection.
4) Utilizing an image enhancement technique based on pixel-level cubic power stretching (CPS) to improve the contrast and visibility of FHB lesions, thereby enhancing the model's ability to differentiate between healthy and diseased regions.

II. Materials and Methods

This section describes the experimental field, data acquisition process, and methodological approach used in this study. Fig. 2 illustrates the complete framework of the proposed study.

A. Experimental Field

In 2024, a total of 9497 wheat breeding lines, along with the parental lines and check varieties, were planted in a single row measuring 1 m in length at Agriculture and Agri-Food Canada's FHB inoculated nursery in Indian

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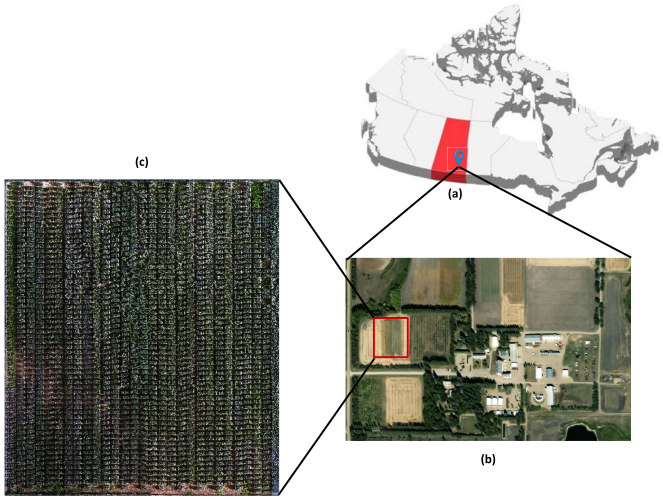


Fig. 3. Overview of the FHB experimental field setup. (a) Location of Indian Head within the province of Saskatchewan, Canada. (b) Aerial view of Indian Head Agricultural Research Foundation (IHARF), where field trials were conducted. (c) Haigh-resolution mosaic image of the FHB experimental field used for data collection and analysis.

Head, Saskatchewan (50.533934, −103.654191). The rows were spaced 30 cm apart, with a border of 0.5 m between each set of six rows. In the nursery, the corn kernel inoculum of *Fusarium graminearum* was used for field inoculation to facilitate the development of FHB disease symptoms. The protocol for the *Fusarium graminearum* corn kernel inoculum preparation was adopted from [52]. Field inoculation with corn inoculum was previously described in [53]. Briefly, the corn inoculum was dispersed at a rate of 8 g per row with two applications at weekly intervals, starting when the plants reached the 4–6 leaf stage.

The pathogen responsible for FHB in wheat is highly influenced by environmental conditions, which play a critical role in both the initiation and progression of the disease. Successful infection and the subsequent development of disease symptoms typically occur when temperatures range between 16 °C and 30 °C under conditions of high relative humidity (RH) [54]. An RH level of 70%–90%, sustained for 48 to 60 h, is important for facilitating initial infection and promoting further disease progression [55]. To ensure effective screening of wheat lines for FHB resistance, specialized disease nurseries are established in hot and humid regions of the Canadian Prairies, such as the facility in Indian Head, Saskatchewan. These nurseries are equipped with overhead sprinkler irrigation systems to supplement natural rainfall and maintain the warm, humid conditions necessary to increase wheat susceptibility and support consistent FHB infection for reliable disease evaluation.

During the data collection period (July and August), the mean daily temperature in Indian Head was 19.4 °C in July and 18.0 °C in August, with a mean RH of 72% (July) and 64% (August), and the total precipitation prior to and during the imaging window was 74.9 mm in June, followed by 37.4 mm in July and 71.8 mm in August [56]. These conditions fall within the 15 °C–25 °C temperature window widely reported as optimal for Fusarium graminearum infection and FHB development, particularly when combined with high humidity and sustained spike wetness near flowering stages [57], [58], [59]. A consistent hot and humid environment, supplemented by overhead sprinkler irrigation three times per week in addition to natural precipitation, successfully created favorable conditions for FHB development. This environment promoted uniform infection symptoms on both the test lines and susceptible check lines, ensuring consistent disease expression across the field. Such controlled conditions were essential for capturing high-quality UAV imagery, enabling accurate image analysis and reliable modeling for FHB detection. The incidence and severity of FHB were assessed with a visual assessment 21 days after inoculation. Fig. 3 provides an overview of the FHB field for clearer visualization.

B. Data Acquisition

This study utilized the DJI Mavic 3T drone to capture RGB images from various heights and angles, ensuring a detailed assessment of FHB symptoms. The Mavic 3T is renowned for its compact and portable design, featuring a 48-Mpixel CMOS wide-angle camera, 56× hybrid zoom, and DJI O3 Enterprise transmission, making it highly effective for precision agriculture applications. Its thermal imaging capabilities, although primarily designed for firefighting and inspections, enhance its versatility in agricultural monitoring. The 12-Mpixel zoom camera, which can achieve up to 56× hybrid zoom, was particularly useful for capturing the fine details of the wheat spikes infected by FHB.

Data collection was carried out over several weeks, targeting 200 selected plots in the field. Flights were conducted between 11:00 A.M. and 2:00 P.M. on sunny days to ensure data reliability, with the camera sensor held at different angles.

1) UAV Imaging Protocol and Parameter Sweep: To determine the optimal settings for FHB spike data acquisition, we conducted a systematic sweep of UAV flight heights, camera/gimbal tilt angles, drone azimuths, and zoom levels. Four tilt strategies were tested (−30°, −45°, −60°, and −75°), and images were captured at heights ranging from approximately 12–55 ft above ground level.

For each plot, the drone was manually rotated through azimuths from 270° to 360° in 15° increments to reduce glare and capture uniform illumination. A fixed optical zoom of 8.5× was used to ensure consistent spatial detail across all captures. Because this study focused on manual, plot-level imaging rather than field-wide surveys or mosaicking, overlap ratios and flight speeds did not apply to our protocol. Each image represents a single-view, high-resolution capture of one plot.

Table I summarizes the tested configurations and highlights the optimal setting, 12–27 feet height, −30° and −45° gimbal tilt, full drone azimuth rotation, and 8.5× zoom, used in final data collection. This configuration provided the best tradeoff between minimizing occlusion, avoiding overexposure, and ensuring high-resolution spike visibility.

2) Impact of UAV Flight Parameters on Image Quality: Our parameter sweep revealed that camera gimbal tilt and flight height have a significant impact on image quality and

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TABLE I
UAV IMAGING STRATEGIES AND FLIGHT PARAMETERS FOR MANUAL PLOT-LEVEL DATA ACQUISITION

Strategy	Camera/Gimbal Tilt (°)	Flight Height Range (feet)	Drone Azimuth/Yaw (°)	Zoom (×)	Description / Purpose
Strategy 1	-30	12 – 18	270 – 360 (15° increments)	8.5	Low oblique view minimizing canopy overlap, providing excellent spike visibility for detection.
Strategy 2	-45	14 – 27	270 – 360 (15° increments)	8.5	Moderate oblique tilt balancing visibility and coverage, ideal for uniform spike imaging and reduced occlusion.
Strategy 3	-60	19 – 35	270 – 360 (15° increments)	8.5	Steeper tilt with reduced occlusion but slight perspective distortion in outer plot areas.
Strategy 4	-75	30 – 55	270 – 360 (15° increments)	8.5	Near-nadir view for complete plot coverage but increased glare and reduced spike clarity.

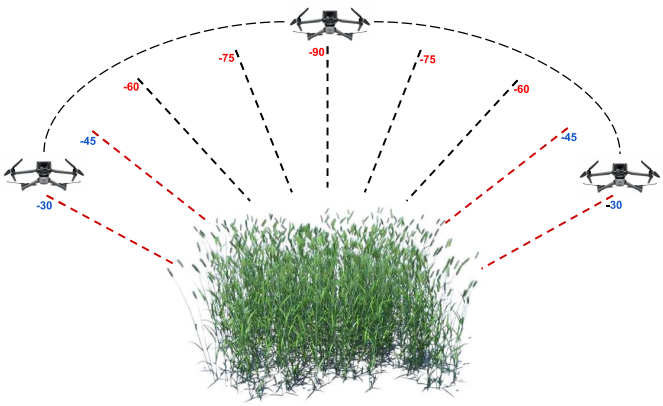


Fig. 4. UAV imaging strategies with four gimbal tilt angles (−30°, −45°, −60°, and −75°) used for manual, plot-level data capture. The red lines indicate the optimal configuration derived from the parameter sweep. This setting minimized occlusion and overexposure while preserving spike-level detail for accurate FHB detection.

downstream detection. Moderate oblique tilts (−30° and −45°) consistently reduced canopy occlusion and produced more uniform, clearly visible FHB spikes, improving contour continuity for annotation and detection. In contrast, steeper tilts (−60° and −75°) increased foreshortening and highlight/shadow artifacts, which obscured spike boundaries and reduced usable detail.

Flight height exhibited a similar tradeoff. Low heights increased within-plot occlusions, while high heights diminished spike-level detail. Moderate heights paired with −30° to −45° tilts delivered the best balance of spike visibility, low occlusion, and stable exposure. This empirically selected operating window, with an oblique tilt at a moderate height, is illustrated in Fig. 4, yielding higher quality annotations and more reliable model performance.

C. Image Enhancement for FHB Detection

Capturing high-quality images of FHB lesions in wheat spikes was challenging due to the high ambient light levels, which often resulted in overexposure in the captured images. The similarity between FHB-infected regions and the surrounding background further complicated the detection process, making it difficult for convolutional neural networks to extract meaningful features. Since color plays a critical role in identifying FHB lesions, an effective solution is the application of pixel-level CPS to adjust the brightness levels of the image. This method expands overexposed (high brightness) regions to recover lost details while applying milder corrections to underexposed (low brightness) areas to retain subtle variations. The adjusted pixel values are then normalized across the image to enhance global contrast, sharpening the distinction between the lesions and their surroundings.

In wheat images, FHB lesions typically exhibit high pixel intensity across the RGB channels. In contrast, background elements such as leaves and soil have relatively lower intensity values. To improve the visibility of these lesions and enhance FHB detection accuracy, we applied CPS to highlight their distinct color characteristics. First, the pixel values in the RGB channels are individually transformed using a cubic power function, as defined in the following equation:

$$\left\{ \begin{array}{l} V_r = r^3(x,y), \\ V_g = g^3(x,y), \\ V_b = b^3(x,y) \end{array} \right\} \tag{1}$$

where $r^3(x,y)$, $g^3(x,y)$, and $b^3(x,y)$ represent the cubic power transformation applied to the RGB pixel values at coordinates (x,y). The resulting stretched values are denoted as V_r, V_g, and V_b. Next, the transformed pixel values undergo normalization, as shown in the following equation:

$$V'_{r,g,b} = \left(\frac{V_{r,g,b}(x,y) - \min(V_{r,g,b}(x,y))}{\max(V_{r,g,b}(x,y)) - \min(V_{r,g,b}(x,y))} \right) \times 255 \tag{2}$$

where $V'_{r,g,b}$ represents the normalized brightness values of the R, G, and B channels and $\min(V_{r,g,b}(x,y))$ and $\max(V_{r,g,b}(x,y))$ correspond to the minimum and maximum pixel values in each respective channel, respectively.

By applying these transformations, the contrast between the FHB lesions and the surrounding background is enhanced, making the infected areas more distinguishable. This improvement enhances the model's ability to extract relevant features, resulting in more accurate FHB detection. The effectiveness

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Fig. 5. Application of CPS to enhance FHB lesion visibility. The top row shows the original images, while the bottom row displays the corresponding CPS-processed images. The CPS algorithm effectively enhances lesion visibility by improving contrast and reducing overexposure, making FHB-infected regions more distinct and easier to identify.

TABLE II
STATISTICAL DISTRIBUTION OF IMAGES AND ANNOTATED FHB SPIKES ACROSS TRAINING, VALIDATION, AND TESTING SUBSETS

Dataset Split	Number of Images	FHB-infected Spikes
Training (80%)	400	18,000 – 20,000
Validation (10%)	50	2,250 – 2,500
Testing (10%)	50	2,250 – 2,500
Total	**500**	**22,500 – 25,000**

of CPS in improving the visibility of the FHB lesion is demonstrated in the processed images shown in Fig. 5.

D. Data Preparation and Annotation

To develop a dataset, we collected data from 200 wheat plots and selected 500 high-quality images for FHB detection. Each selected image was divided into four equal slices to facilitate easier labeling and detection.

The primary focus of this study was to develop a model that exclusively detects FHB-infected wheat spikes. Therefore, only diseased spikes were manually annotated, while healthy spikes were not labeled and were treated as background during training. Across the dataset, each image contained an average of 45–50 diseased spikes, resulting in a total of approximately 22 500–25 000 annotated spikes.

To ensure unbiased evaluation and prevent data leakage, the dataset was split into training (80%), validation (10%), and testing (10%) subsets. A stratified random sampling strategy was used, where each subset contained images drawn from different field plots to maintain diversity and avoid overfitting. Table II summarizes this distribution across the train, validation, and test subsets, ensuring a balanced dataset for model training and evaluation.

Angle coverage and evaluation design for the training, validation, and test sets comprise images captured at −30° and −45° gimbal tilts across multiple heights. These angles were selected after preliminary trials indicated that steeper tilts introduced foreshortening and highlight/shadow artifacts that obscured spike boundaries. To analyze the impact of acquisition height and angle on detection, we: 1) perform an in-distribution analysis across height bins within −30°/−45° and 2) conduct a robustness check on a small set of −60°/−75° images to quantify the expected domain shift, noting that the model was not trained on these angles.

In UAV-based remote sensing images, wheat spikes appear at various angles and orientations due to the top-down perspective of image acquisition. This variation makes it difficult to detect and annotate FHB-infected spikes. To address this, we used the OBB method for annotation using the computer vision annotation tool. Unlike standard AABB, which may include unnecessary background areas and reduce annotation precision, OBB provides a more precise representation by aligning with the actual shape and orientation of the spikes. This approach ensures more accurate localization of FHB-infected spikes, thereby improving the accuracy of the detection model.

E. Methodology

We adopt YOLOv10 and YOLOv11 as our baseline models because they strike an optimal balance between detection accuracy and computational efficiency, making them suitable for detecting FHB. In this section, we first provide detailed descriptions of the YOLOv10 and YOLOv11 architectures. We then introduce our improved hybrid model, which further integrates the strengths of both architectures to enhance detection accuracy and efficiency for OBB detection.

1) Baseline Architectures: YOLOv10 advances real-time object detection through a combination of architectural innovations and training strategies that remove the need for postprocessing, such as nonmaximum suppression (NMS). YOLOv10 employs a dual-branch training strategy. One branch uses a one-to-many assignment to provide rich supervisory signals, while the second branch uses one-to-one matching optimized for inference. A rank-guided block design replaces redundant components with more compact alternatives, e.g., a CIB, to achieve the best tradeoff between efficiency and detection performance. The matching metric is defined in (3). SCDown separates channel expansion (via point-wise convolutions) and spatial reduction (via depth-wise convolutions), reducing both parameter count and latency. A lightweight classification head reduces computational overhead without sacrificing accuracy. YOLOv10 leverages large-kernel convolutions in deeper layers and integrates a partial self-attention (PSA) module. These modifications expand the receptive field and enhance the model's ability to represent global features while maintaining a low computational cost

$$m(\alpha, \beta) = s \cdot p^{\alpha} \cdot \mathrm{IoU}(\hat{b}, b)^{\beta} \tag{3}$$

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where s represents the spatial prior indicating whether the anchor point of prediction is within the instance, p is the classification score; IoU denotes the intersection over union between the predicted box $\hat{b}$ and the ground-truth box b; and α and β are important hyperparameters that balance the impact of the semantic prediction task and the location regression task.

Building on the legacy of previous YOLO iterations, YOLOv11 introduces several architectural improvements designed to refine feature extraction and broaden the range of supported computer vision tasks. YOLOv11 replaces traditional bottleneck structures with the novel C3k2 block. This design increases speed and reduces the parameter count while preserving accuracy. The architecture integrates the spatial pyramid pooling-fast (SPPF) block with an additional C2PSA module. The C2PSA block introduces spatial attention, allowing the network to focus on critical regions and improving the detection of small or occluded objects. Unlike YOLOv10, YOLOv11 relies on traditional NMS postprocessing but achieves superior accuracy through architectural refinements, including DWConv in detection heads for lightweight classification. The head processes the aggregated features with multiple convolution-batchNorm-siLU layers to produce bounding-box predictions and class scores.

2) Modified Hybrid YOLO Model: Our proposed modified hybrid YOLO model is specifically designed for OBB detection, enabling the accurate identification of FHB spikes in UAV-based imagery. This model combines key innovations from YOLOv10 and YOLOv11 to overcome their limitations. YOLOv10 is highly efficient due to its CIB module and rank-guided design strategy. However, it is primarily designed for AABB detection and does not support OBB detection. This is a major drawback in agricultural applications, where wheat spikes appear at various angles and require precise orientation-aware detection.

On the other hand, YOLOv11 introduces OBB detection, making it more suitable for detecting rotated objects. However, it struggles with small and densely packed targets, which is a significant challenge in UAV imagery where wheat spikes appear as tiny, densely clustered objects. Furthermore, in the context of FHB detection, YOLOv11 struggles to distinguish between diseased spikes and visually similar plant structures, such as leaf structures that visually resemble FHB lesions, often leading to false positives (FPs) and compromised precision. To better address the dual challenges, we developed a modified hybrid YOLO model that achieves computational efficiency while effectively detecting small and densely packed oriented FHB wheat spikes in UAV imagery. Our modified hybrid YOLO model is illustrated in detail in Fig. 6.

In our architecture, the backbone incorporates a dual strategy. We adopt the YOLOv11-inspired C3k2 block, which employs dual sequential smaller convolutions (kernel size 2) instead of a single large kernel, enhancing fine-grained feature capture while reducing computational overhead. To address redundancy in deeper layers, we replace standard bottlenecks with C2fCIB blocks adapted from YOLOv10 at critical stages (blocks 6 and 8 in the backbone and 16 and 19 in the neck). The C2fCIB block fuses the traditional C2f design with a CIB, prioritizing parameter efficiency.

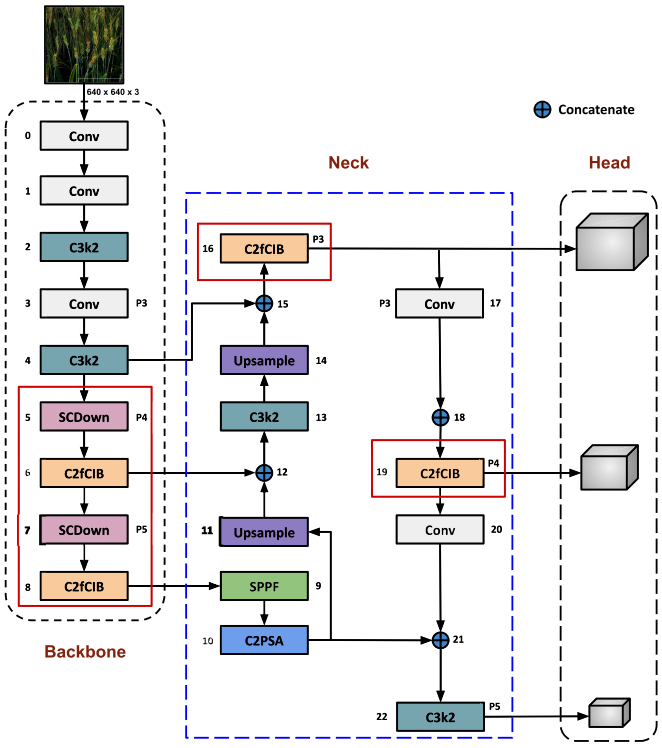


Fig. 6. Overview of the modified hybrid YOLO architecture for OBB detection of FHB UAV-based wheat spikes.

TABLE III
C2FCIB BLOCK USAGE ACROSS MODEL VARIANTS

Model Variant	C2fCIB Blocks
n	22
s	8, 22
m	8, 19, 22
l	8, 13, 19, 22
x	6, 8, 13, 19, 22

For downsampling, we integrate YOLOv10's SCDown at backbone stages 5 and 7. This module separates spatial reduction through depthwise convolutions from channel expansion through pointwise convolutions, minimizing information loss and reducing GFLOPs compared to conventional strided convolutions. The detection head retains YOLOv11's architecture, leveraging its DWConv design to reduce latency and parameter counts during inference. The training pipeline follows YOLOv11's methodology, utilizing its native label assignment strategy and NMS for postprocessing. This ensures robust bounding-box refinement and precise orientation estimation, which are both critical for OBB tasks. By combining YOLOv11's efficient detection head with YOLOv10's optimized backbone and neck components, our hybrid model achieves a reduction in parameters and GFLOPs compared to YOLOv8 and YOLOv11 while maintaining state-of-the-art accuracy on complex agricultural detection tasks.

a) C2fCIB Block: YOLOv10's stage component employs either a C2f or a modified C2fCIB block. In larger models and deeper stages, redundancy becomes more pronounced,

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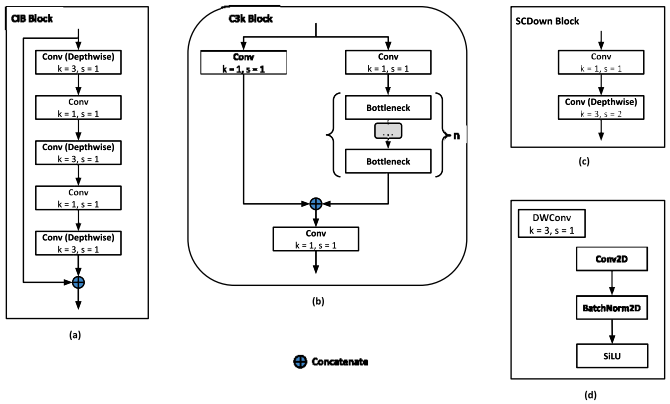


Fig. 7. Key architectural components of our hybrid YOLO model. (a) CIB block that replaces standard bottlenecks in high-redundancy stages, improving efficiency by reducing parameters while maintaining feature integrity. (b) C3k block: a refined convolutional structure consisting of three convolutional layers and multiple bottlenecks, designed to enhance feature extraction with optimized shortcut connections. (c) SCDown block that separates spatial reduction and channel expansion, minimizing computational overhead. (d) A DWConv block integrated into the detection head, reducing parameter count while maintaining detection accuracy.

prompting the replacement of the bottleneck block within C2f with a CIB to form C2fCIB. Table III details the usage of C2fCIB across YOLOv10 variants, whereas our approach consistently applies the C2fCIB block (6, 8, 16, and 19) for all variants. For example, in the YOLOv10s version, C2fCIB is implemented in blocks 8 and 22. This substitution reduces redundancy, enhances model efficiency, and accelerates training. Some CIB implementations do not use shortcut connections, while others do. Our hybrid model adopts shortcuts (shortcut = true) in all CIB blocks to improve the information flow. The CIB block is reported in Fig. 7(a).

b) C3k2 Block: In YOLOv11, the stage component incorporates the C3k2 block, which is an improved version of the C2f block used in YOLOv8 and YOLOv10. This new block reduces the number of parameters while enhancing efficiency. A key element of C3k2 is the "c3k" parameter, which determines whether the module activates the C3k variant or falls back to a traditional bottleneck design as seen in C2f. When c3k is set to true, the module employs the C3k block, a structure comprising three convolutional blocks and several bottlenecks with options for shortcuts or no shortcuts (in our case, shortcut = true). In the original YOLOv11 model, the c3k setting is always true for the *m*, *l*, and *x* variants; however, in our hybrid model, we enforce c3k as true for all variants, such as nano, small, medium, large, and extralarge (*n*, *s*, *m*, *l*, and *x*), to improve feature extraction. This design enables the C3k2 block (2, 4, 13, and 22) to capture more complex features, making it particularly effective for objects of varying sizes (see Fig. 7(b) for an illustration of the C3k block).

c) SCDown Block: YOLOv8 and YOLOv11 use standard 3×3 convolutions with a stride of 2 for downsampling, a conventional approach that can be resource-intensive due to its high computational cost and large parameter count. In contrast, YOLOv10 introduces SCDown, which separates spatial reduction from channel expansion. SCDown first adjusts the channel dimensions using a 1×1 pointwise convolution, keeping the spatial resolution intact, and then reduces the spatial resolution with a 3×3 depthwise convolution without changing the number of channels. This method significantly decreases both computational demands and the number of parameters. Our hybrid model adopts the SCDown block (5 and 7) in place of the traditional convolutional block to enhance the efficiency. The SCDown block is depicted in Fig. 7(c).

d) SPPF and C2PSA: Our hybrid model incorporates SPPF to enable multiscale representation of feature maps. By pooling at different scales, SPPF captures features at various levels of abstraction. Originally introduced in YOLOv8 and later adapted for YOLOv10 and YOLOv11, SPPF is an enhanced version of the earlier spatial pyramid pooling (SPP) block. Unlike SPP, which utilizes maxpool kernels of sizes 3, 5, and 9, SPPF employs a consistent kernel size of 5, significantly reducing the number of floating-point operations required. Moreover, our model leverages an advanced attention mechanism initially introduced in YOLOv10 and refined in YOLOv11, where it is implemented as the C2PSA block. This block features two convolutional stages, an initial and a terminal stage. The output from the initial convolution is split into two parts; one part is processed through multiple PSABlocks to determine the most informative features, while the other is later combined via a concatenation block. Each PSABlock consists of an attention module alongside a feedforward network (FFN), with the FFN comprising two consecutive convolutional layers—the second of which omits an activation function to streamline processing.

e) DWConv Head Block: In our hybrid model, we incorporate the YOLOv11 detection block, which features depthwise convolutions in its classification head, a design that closely mirrors that of YOLOv10. However, YOLOv11 distinguishes itself by integrating a novel DWConv component. We employ the C2fCIB block responsible for detecting small and medium objects, while the C3k2 block is dedicated to the detection of large objects. The DWConv block is presented in Fig. 7(d).

In general, by merging the advanced feature extraction capabilities of YOLOv11's C3k2 block with the efficiency improvements and redundancy reduction of YOLOv10's C2fCIB block, our hybrid YOLO model offers an optimized solution for OBB detection. The combined use of SCDown, a DWConv-based detection head, and a dual strategy incorporating both NMS-free and NMS-based mechanisms enables superior detection accuracy and computational efficiency. This integrated architecture is effective for real-time detection in agricultural scenarios, achieving lower latency and resource usage compared to models such as YOLOv8-OBB and YOLOv11-OBB while precisely localizing and orienting wheat spikes from UAV imagery. To improve clarity and readability, Table IV summarizes the architectural similarities and differences between YOLOv10, YOLOv11, and our proposed Hybrid YOLO-OBB model.

F. Evaluation Metrics

The evaluation of the models was conducted using key metrics, including precision (*P*), recall (*R*), and average precision

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TABLE IV
COMPARISON OF YOLOV10, YOLOV11, AND THE PROPOSED HYBRID YOLO-OBB MODEL

Model	Core Building Blocks	Downsampling Strategy	Detection Head	Key Features / Modules	Primary Design Focus
YOLOv10	CIB	SCDown	Standard YOLO Head	Reduced parameters and FLOPs, high efficiency	Lightweight and Real-Time Inference
YOLOv11	C3k2 Blocks	Standard Downsampling	DWConv-based Head	Improved orientation handling and feature extraction	High Detection Accuracy
Hybrid	Combination of CIB (YOLOv10) + C3k2 (YOLOv11)	SCDown + Standard Downsampling (Integrated)	DWConv	Combines efficiency of YOLOv10 with orientation-aware features of YOLOv11	Balanced Accuracy and Speed with OBB Support

(AP). These metrics offer insights into the model's performance by analyzing its ability to identify positive instances and avoid incorrect predictions. P measures the proportion of predicted positive instances that are correct. In other words, it reflects the accuracy of the model's positive predictions. R, on the other hand, quantifies the fraction of actual positive instances that the model successfully identifies. It indicates how well the model can capture all relevant cases. Both P and R are computed based on four fundamental components: true positives (TPs), true negatives (TNs), FPs, and false negatives (FNs). These components form the basis for assessing the tradeoffs between accurate detections and missed or incorrect predictions. The IoU metric is used to evaluate the spatial alignment between predicted bounding boxes and ground-truth bounding boxes. IoU is calculated by dividing the area of overlap between the predicted box (Pr) and the ground-truth box (Gr) by the total area covered by both boxes. This metric provides a quantitative measure of how closely the predicted box matches the true object location, with higher IoU values indicating greater similarity between the two boxes

$$P = \frac{\text{TP}}{\text{TP} + \text{FP}}$$

$$R = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

$$\text{IoU}(\text{Gr}, \text{Pr}) = \frac{\text{Gr} \cap \text{Pr}}{\text{Gr} \cup \text{Pr}}. \tag{4}$$

A high P score indicates that the model produces very few FP predictions, meaning that most of the predicted positive instances are indeed correct. On the other hand, a high R score indicates the model's ability to accurately detect the majority of ground-truth objects, resulting in minimal missed detections. The AP is calculated as the area under the P–R curve for a specific class. It provides a balanced and comprehensive assessment of the tradeoffs between P and R. To evaluate the overall performance across all classes, the mean AP (mAP) is calculated by averaging the AP values obtained for each class

$$\text{mAP} = \frac{1}{N} \sum_{i=1}^{N} \text{AP}_i. \tag{5}$$

III. RESULTS AND DISCUSSION

This section discusses our experimental setup and presents the results obtained using our modified hybrid model for OBB detection on UAV-based wheat spike imagery. We compared and discussed our model results with state-of-the-art OBB detectors. We evaluated and compared these detectors using our custom UAV-based OBB FHB dataset.

A. Training and Parameter Setting

In our experiments, the modified hybrid YOLO model was trained using Python 3.10.15 and Torch 2.5.1 on an NVIDIA GeForce RTX 4070 Ti SUPER GPU with 16 GB of VRAM. The training setup consisted of 400 epochs, with an input resolution of 640×640 pixels and a batch size of 16, and an early stopping patience of 30 epochs. An automatic optimizer selection process determined the use of AdamW with an initial learning rate of 0.002, a momentum of 0.9, and a weight decay of 0.0005, optimized across multiple parameter groups. Key enhancements, including mixed-precision training (AMP enabled) and robust data augmentation strategies such as mosaic augmentation and random augmentation, were incorporated to accelerate convergence and improve generalization for OBB detection.

B. Comparison With State-of-the-Art YOLO-OBB Models

We evaluate our proposed Hybrid YOLO against YOLOv8-OBB and YOLOv11-OBB across all model variants (n, s, m, l, and x) on a custom FHB dataset. The analysis focuses on parameter efficiency, computational complexity, precision–recall balance, and mAP under two IoU thresholds: 0.5 (strict localization) and 0.5–0.95 (generalized detection).

1) Quantitative Results: In terms of parameter efficiency, our model outperformed its counterparts. For instance, the Hybrid-YOLO-n variant achieved 2.3 M parameters, representing a 25.8% reduction compared to YOLOv8n-OBB (3.1M) and a 14.8% reduction compared to YOLOv11n-OBB (2.7 M). This reduction is attributed to the integration of C2fCIB blocks and depthwise convolutions, which replace redundant operations in the backbone and neck. Scaling to larger variants, Hybrid-YOLO-x uses 48.4 M parameters, 30.4% fewer than YOLOv8x-OBB (69.5 M) and 17.7% fewer than YOLOv11x-OBB (58.8 M). Our model also demonstrated computational efficiency. For example, Hybrid-YOLO-s operates at 21.6 GFLOPs, a 27% reduction compared to YOLOv8s-OBB (29.6 GFLOPs) and a 4% reduction compared to YOLOv11s-OBB (22.5 GFLOPs). This efficiency stems from the SCDown module and C3k2 blocks, which minimize spatial–channel redundancy. For large-scale detection, Hybrid-YOLO-l requires 84.8 GFLOPs, 49.8% lower than

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TABLE V
PERFORMANCE COMPARISON OF OUR PROPOSED HYBRID YOLO WITH YOLOV8-OBB AND YOLOV11-OBB FOR ALL VARIANTS ON THE CUSTOM FHB DATASET

Model	Parameters (M) ↓	GFLOPs ↓	Precision (%)	Recall (%)	$mAP^{val}_{0.5}$ ↑	$mAP^{val}_{0.5-0.95}$ ↑	Performance Efficiency Score ↑ ($mAP_{0.5-0.95}$/GFLOPs)
YOLOv8n-OBB	3.1	8.4	**79.5**	80.1	88.1	58.7	6.99
YOLOv11n-OBB	2.7	6.7	73.4	**90.7**	88.5	60.2	8.98
Hybrid-YOLO-n	**2.3**	**6.5**	75.9	87.5	**89.6**	**60.3**	**9.28**
YOLOv8s-OBB	11.4	29.6	82.7	76.8	90.0	58.9	1.99
YOLOv11s-OBB	9.7	22.5	83.9	**85.4**	91.4	63.6	2.83
Hybrid-YOLO-s	**8.2**	**21.6**	**85.6**	84.4	**93.0**	**66.3**	**3.07**
YOLOv8m-OBB	26.4	81.2	84.2	80.4	90.2	61.5	0.76
YOLOv11m-OBB	20.9	71.9	81.4	**85.9**	91.7	62.3	0.87
Hybrid-YOLO-m	**16.9**	**67.7**	**87.8**	84.4	**93.2**	**66.4**	**0.98**
YOLOv8l-OBB	44.5	169.1	76.2	**88.3**	90.5	63.1	0.37
YOLOv11l-OBB	26.2	91.0	**86.1**	82.4	91.8	64.4	0.71
Hybrid-YOLO-l	**21.6**	**84.8**	85.0	86.2	**93.5**	**67.1**	**0.79**
YOLOv8x-OBB	69.5	263.9	86.2	80.9	90.7	63.4	0.24
YOLOv11x-OBB	58.8	203.8	**87.5**	81.8	92.5	67.1	0.33
Hybrid-YOLO-x	**48.4**	**189.3**	85.9	**87.8**	**93.8**	**67.7**	**0.36**

YOLOv8l-OBB (169.1 GFLOPs), highlighting its scalability for UAV deployments.

The hybrid YOLO-s variant achieves the most substantial relative improvement with a precision of 87.8% compared to 83.9% for YOLOv8s-OBB and 82.7% for YOLOv11s-OBB. Similarly, our small variant reaches an mAP@0.55of 93.8%, outperforming both YOLOv8s-OBB (91.4%) and YOLOv11s-OBB (90.0%). The most notable improvement is observed in the mAP@0.5–0.95 metric, where hybrid-YOLO-s achieves 66.3%, representing a 7.4 point improvement over YOLOv8s-OBB and 2.7 points over YOLOv11s-OBB. For the larger model variants, Hybrid-YOLO-*l* achieves a balanced performance with 85.0% precision and 85.9% recall, leading to an mAP@0.5value of 93.6% and mAP@0.5–0.95 of 67.1%. This represents a substantial improvement over YOLOv8l-OBB's 90.5% mAP@0.5and 63.1% mAP@0.5–0.95. Similarly, our extralarge variant (Hybrid-YOLO-*x*) achieves the highest mAP@0.5–0.95 score of 67.7%, surpassing both YOLOv8x-OBB (60.0%) and YOLOv11-OBBx (67.4%). Table V presents the detailed comparison results of our proposed hybrid YOLO with the YOLOv8-OBB and YOLOv11-OBB models.

2) Qualitative Results: Visual inspection of the detection outputs reveals significant performance differences between our proposed hybrid YOLO model and existing state-of-the-art YOLO-OBB models. Fig. 8 provides a side-by-side comparison that illustrates these differences across representative UAV field images.

The YOLOv8-OBB model exhibits several noticeable limitations in FHB spike detection. First, it frequently misclassifies leaf structures as FHB lesions when they share visual similarities with infected spikes. This confusion between plant parts reduces precision and introduces FPs, which can compromise subsequent disease assessments. Furthermore, YOLOv8-OBB exhibits duplicate detections, generating multiple overlapping bounding boxes for a single FHB-infected spike, which leads to redundant detections. Another shortcoming is its inability to discriminate between healthy and FHB-infected spikes, resulting in incorrect detection of non-FHB spikes as diseased.

The YOLOv11-OBB model shows improvement over YOLOv8-OBB by largely addressing the overprediction issue; however, it still struggles with the misclassification of leaves that resemble FHB lesions. Similar to YOLOv8-OBB, YOLOv11-OBB also predicts some non-FHB spikes as infected, which reduces its overall reliability in real-world agricultural applications.

In contrast, our hybrid YOLO model demonstrates improvements in detection accuracy and robustness. It does not suffer from overprediction or misclassification of leaves as FHB lesions. The model accurately identifies FHB spikes with high precision, even in complex scenarios where leaves and spikes overlap or have a similar appearance. Additionally, the hybrid model effectively distinguishes between healthy and diseased spikes, minimizing FPs and ensuring reliable detection.

C. Comparison With Other State-of-the-Art OBB Detection Models

We further evaluated our hybrid YOLO model against a diverse set of state-of-the-art-oriented object detection approaches beyond the YOLO family. As presented in Table VI, our model achieves an mAP of 93.8%, outperforming all compared methods, including Faster *R*-CNN OBB (87.7%), RDFPN (89.2%), R^2CNN (89.9%), RRPN (90.4%), ROI-Transformer (92.1%), and S^2ANet (92.4%).

The performance advantages stem from our model's ability to capture the unique characteristics of FHB-infected wheat spikes effectively. Traditional two-stage detectors, such as Faster *R*-CNN OBB, while robust for general object detection, lack the specialized architectural components necessary for the fine-grained detection of elongated agricultural objects, like wheat spikes. Similarly, while RRPN and ROI-Transformer

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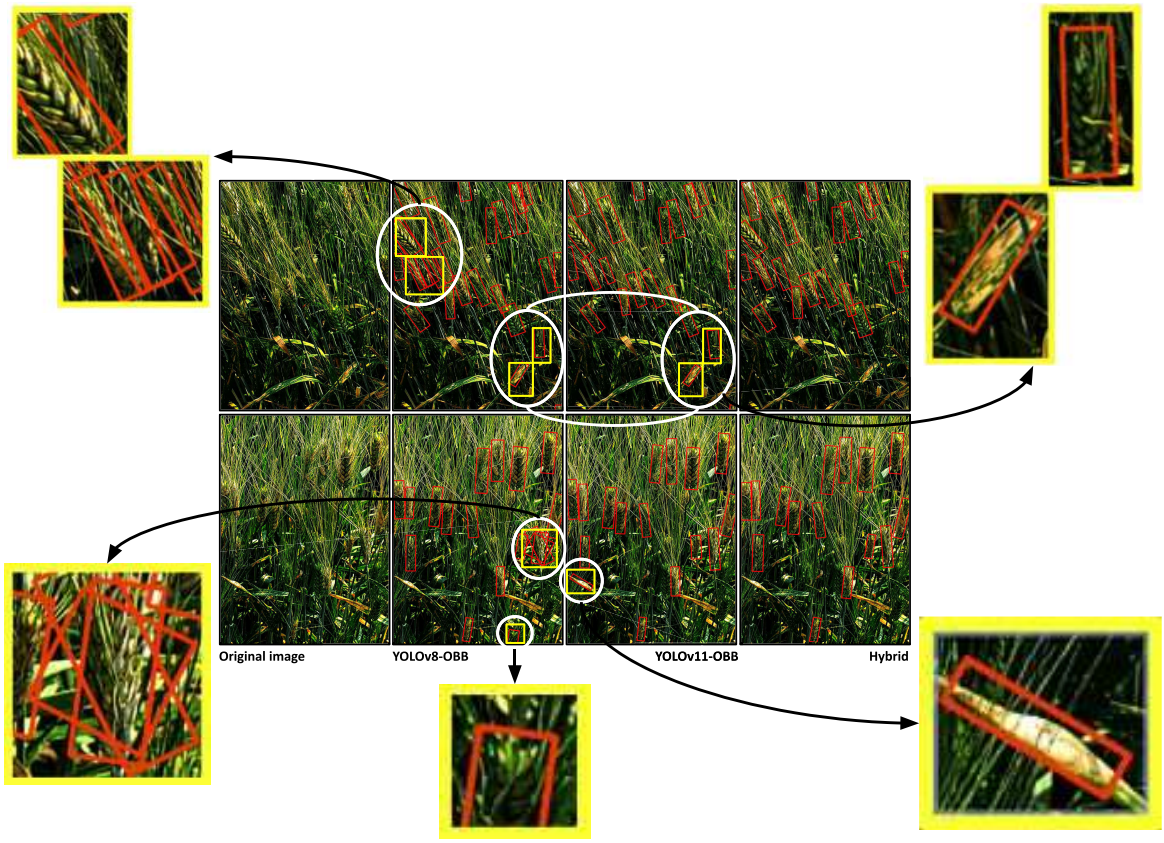


Fig. 8. Visual comparison of FHB spike detection results across representative UAV field images using YOLOv8-OBB, YOLOv11-OBB, and the proposed hybrid YOLO model.

TABLE VI

COMPARISON WITH OTHER STATE-OF-THE-ART OBB DETECTION MODELS ON THE CUSTOM FHB DATASET. FR-O DENOTES THE FASTER *R*-CNN OBB DETECTOR. RDFPN REPRESENTS ROTATION-DENSE FEATURE PYRAMID NETWORKS, WHICH ALSO EMPLOY ROTATED ANCHORS ALONG WITH A MODIFIED FPN DESIGN. R^2CNN STANDS FOR ROTATIONAL REGION CNN. RRPN REFERS TO ROTATION REGION PROPOSAL NETWORKS THAT UTILIZE ROTATED ANCHORS. ROI TRANSFORMER DENOTES THE REGION OF INTEREST TRANSFORMER, AND FINALLY, S^2ANET IS A SINGLE-SHOT ALIGNMENT NETWORK

Model	Parameters (M) ↓	GFLOPs ↓	mAP ↑	Performance Efficiency Score ↑ (mAP/GFLOPs)
FR-O [60]	60.4	172.1	87.7	0.51
RDFPN [61]	-	-	89.2	-
R^2CNN [62]	60.2	168.1	89.9	0.54
RRPN [47]	-	-	90.4	-
RoI-Transformer [63]	74.1	200.0	92.1	0.46
S^2ANet [64]	55.5	198.2	92.4	0.47
Hybrid-YOLO-l	**21.6**	**84.8**	**93.5**	**1.10**

TABLE VII

IMPACT OF UAV FLIGHT HEIGHT AND CAMERA TILT ANGLE ON FHB SPIKE DETECTION PERFORMANCE

Tilt Angle (°)	Flight Height (feet)	Average mAP@0.5 (%)	Average Precision (%)	Average Recall (%)
-30	12-15	94.2	91.8	92.5
	16-18	92.9	90.9	92.2
-45	14-20	92.6	90.4	91.8
	21-27	91.5	89.9	91.3
-60	19-27	62.4	58.8	60.1
	28-35	59.7	56.4	57.9
-75	30-42	58.5	54.8	56.5
	43-55	55.1	52.6	53.2

introduced rotated anchors and specialized ROI operations, their general-purpose design does not fully address the challenges specific to FHB detection in field conditions.

S^2ANet, being the closest competitor with an mAP of 92.4%, demonstrates the effectiveness of single-shot alignment networks for oriented object detection. However, our Hybrid YOLO model's integration of enhanced feature extraction and orientation-specific components provides the marginal improvement needed to push performance to 93.8% mAP, establishing a new state-of-the-art for this application.

D. Impact of Flight Heights and Angles

Performance varied with height for both −30° and −45° tilts. Moderate heights paired with oblique tilts yielded the best balance between spike visibility and occlusion, producing the highest mAP values.

Evaluating the trained model on −60°/−75° images (unseen during training) led to a measurable drop in mAP, consistent with angle-induced domain shift (foreshortening and glare).

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TABLE VIII
ABLATION STUDY: PERFORMANCE COMPARISON OF CONV BLOCK ACROSS MODEL VARIANTS

Variant	Parameters (M)	GFLOPs	Precision (%)	Recall (%)	$mAP^{val}_{0.5}$	$mAP^{val}_{0.5-0.95}$
n	3.1	7.9	78.9	79.1	86.6	58.4
s	10.7	26.0	80.0	83.9	90.8	59.6
m	25.3	83.8	83.3	82.8	91.8	62.9

TABLE IX
ABLATION STUDY: PERFORMANCE COMPARISON OF SCDOWN BLOCK ACROSS MODEL VARIANTS

Variant	Parameters (M)	GFLOPs	Precision (%)	Recall (%)	$mAP^{val}_{0.5}$	$mAP^{val}_{0.5-0.95}$
n	2.7	7.5	79.2	85.2	89.3	59.6
s	9.1	24.4	75.6	89.3	91.3	61.0
m	21.1	78.6	85.5	83.4	91.9	63.4

These results confirm our acquisition choice (−30°/−45°) and indicate that targeted fine-tuning or tilt-aware augmentation would be needed for cross-angle generalization. The effect of different UAV flight heights and gimbal tilt angles on FHB spike detection performance is summarized in Table VII.

E. Ablation Study

To evaluate the architectural design choices in our proposed hybrid YOLO model, we conducted a comprehensive ablation study focusing on the impact of different computational blocks on detection performance. Following the pattern established in YOLOv10, we systematically varied the placement and type of specialized blocks across model variants to identify optimal architectural configurations. The ablation study investigated three key components.

1) C2fCIB block placement at different network depths.
2) Comparison of conventional Conv blocks against SCDown blocks.
3) Use of C3k2 blocks as in our hybrid model to investigate the comparison of the first two components.

1) C2fCIB Block Placement Strategy: Our starting point was the pattern of C2fCIB block usage across YOLOv10 variants, as detailed in Table III. This pattern shows a strategic increase in the number and position of C2fCIB blocks as the model's capacity increases. We focused our ablation study on the *n*, *s*, and *m* variants, as these configurations differed from the baseline architecture of our hybrid model. Our hybrid model incorporated similar block placements for larger variants (*l* and *x*). For each variant (*n*, *s*, and *m*), we performed a series of controlled experiments. First, we implemented the exact C2fCIB block placement as specified in Table III to establish a reference point aligned with the architecture of YOLOv10. Next, we used conventional Conv blocks throughout the entire architecture, allowing us to quantify the specific performance contribution of C2fCIB blocks. Finally, SCDown blocks were placed at positions 5 and 7 while maintaining the variant-specific C2fCIB block placement at other positions.

For each configuration, we evaluated multiple performance metrics, including parameter count, computational complexity (GFLOPs), *P*, *R*, and mAP scores, as presented in Tables VIII and IX. This comparative analysis reveals the quantitative impact of different block types across model variants. The results demonstrate consistent patterns in how block selection affects both computational efficiency and detection accuracy. Notably, SCDown blocks at positions 5 and 7 consistently reduced computational requirements while maintaining competitive detection performance across all tested variants. We observed parameter reductions ranging from 12.9% to 16.6% and GFLOPs reductions between 5.1% and 6.2% across *n*, *s*, and *m* model variants.

For the *n* variant, SCDown blocks demonstrated an improvement in *R*, increasing from 79.1% to 85.2%, while maintaining a consistent *P* around 79% and showing a marginal increase in mAP@0.5from 86.6% to 89.3%. The s variant exhibited more changes, with SCDown blocks boosting *R* from 83.9% to 89.3%, albeit at the cost of a *P* reduction from 80.0% to 75.6%, yet still achieving a slight improvement in mAP@0.5from 90.8% to 91.3%. In the m variant, the performance metrics showed a more balanced transformation, with SCDown blocks improving *P* from 83.3% to 85.5%, maintaining a comparable *R* of 83.4%, and marginally improving mAP@0.5from 91.8% to 91.9%. This systematic approach allowed us to precisely isolate the impact of block selection and placement on both efficiency and detection performance across different model capacities.

2) Architectural Modifications: Beyond block type selection, we investigated additional architectural modifications that could potentially impact model performance. A key focus was the replacement of the CIB with the traditional bottleneck block in the C2f module across all model variants. This modification increased model complexity, with a rise in both parameter count and computational requirements (GFLOPs) compared to our original CIB-based architecture.

Despite the increased computational overhead, the evaluation metrics showed minimal variations. The *P*, *R*, and mAP scores remained consistent across both configurations, suggesting the robustness of our proposed hybrid architecture. We further explored the impact of shortcut connections by systematically testing scenarios where shortcut connections were set to false (compared to our default setting of true). However, these modifications failed to yield any significant performance improvements, reinforcing the effectiveness of our original architectural design.

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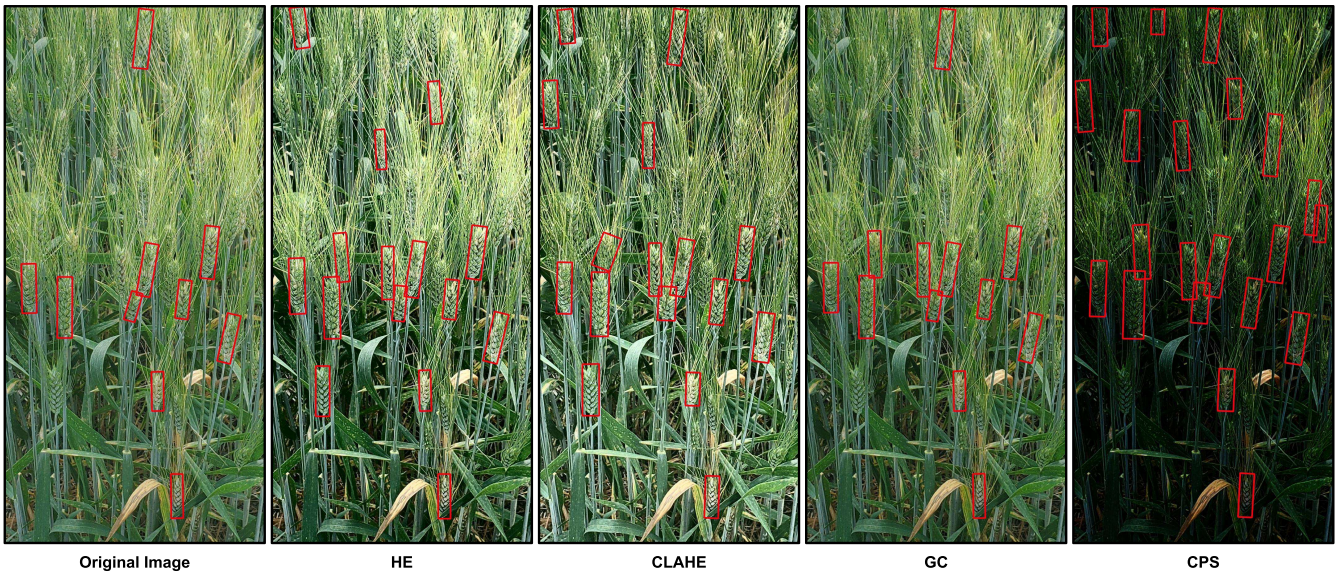


Fig. 9. Detection results using different enhancement techniques for the original image, HE, CLAHE, GC, and CPS. CPS provides the clearest lesion visibility and the most accurate detections.

3) Evaluation of Image Enhancement Techniques: The effect of different image enhancement methods on FHB lesion visibility and detection accuracy was evaluated using a representative UAV image processed through five enhancement pipelines: the original image (no enhancement), histogram equalization (HE), contrast-limited adaptive histogram equalization (CLAHE), gamma correction (GC), and the proposed CPS. As shown in Fig. 9, the original image provided clear wheat spike structures but limited lesion contrast, resulting in missed detections and lower overall accuracy. HE improved global contrast but introduced excessive brightness variations, causing FPs in dense canopy regions. CLAHE provided better localized enhancement and reduced oversaturation, though some patchy artifacts persisted in uniform areas. GC moderately enhanced subtle lesion features but was less effective under variable lighting conditions.

Among all methods, CPS provided the most distinct lesion boundaries and stable visual enhancement, enabling the Hybrid YOLO-OBB model to achieve the highest detection accuracy. CPS minimized FPs while capturing nearly all true lesion instances, even in challenging areas with overlapping spikes. These results confirm CPS as the most suitable preprocessing approach for UAV-based FHB detection.

IV. Conclusion

In this study, we proposed a hybrid YOLO-based detection model that integrates the efficiency of YOLOv10 with the orientation-aware capabilities of YOLOv11 to accurately detect FHB-infected wheat spikes using UAV imagery. The proposed model achieved superior detection accuracy while maintaining a lower computational cost compared to conventional YOLO variants and other state-of-the-art OBB detectors.

The model's design allows it to be adapted to diverse wheat-growing regions by retraining or fine-tuning on localized datasets, accommodating variations in crop type, canopy structure, and environmental conditions such as lighting or soil background. Furthermore, by training on images captured at different growth stages and severity levels, the hybrid model can be extended to detect early stage infections, providing a critical window for timely fungicide application and management decisions. This adaptability highlights the potential for broader use beyond the studied fields, supporting precision agriculture and sustainable disease management strategies.

In future work, we plan to expand this research beyond detection to focus on FHB severity assessment at both the plot level and individual spike level. This task is currently labor-intensive, time-consuming, and highly subjective, as it relies on manual field inspections by trained personnel. Automating this process through advanced UAV-based DL will enable rapid, consistent, and scalable disease assessments across large field trials. Furthermore, we will incorporate multiseason, multiregional datasets with varying disease severities to improve the model's generalization and robustness. These enhancements will support the integration of our framework into precision agriculture pipelines, promoting sustainable wheat production and more effective disease management strategies at a global scale.

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